

Diabetes

November 5, 2025

```
[1]: import numpy as np
import pandas as pd
```

```
[2]: from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, \
recall_score, f1_score
```

```
[3]: df = pd.read_csv("diabetes.csv")
```

```
[4]: df.head(5)
```

```
[4]:
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

	Pedigree	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

```
[6]: print(df.head())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
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	Pedigree	Age	Outcome
0	0.627	50	1
1	0.351	31	0

2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

```
[7]: df.isnull().sum()
```

```
[7]: Pregnancies      0
      Glucose          0
      BloodPressure    0
      SkinThickness    0
      Insulin          0
      BMI              0
      Pedigree         0
      Age              0
      Outcome          0
      dtype: int64
```

```
[8]: print(df.shape)
```

```
(768, 9)
```

```
[9]: # Separate input features (X) and target label (y)
      X = df.drop(columns=['Outcome'])
      X
```

```
[9]:
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	
..	...	...	...	...	...	...	
763	10	101	76	48	180	32.9	
764	2	122	70	27	0	36.8	
765	5	121	72	23	112	26.2	
766	1	126	60	0	0	30.1	
767	1	93	70	31	0	30.4	

	Pedigree	Age
0	0.627	50
1	0.351	31
2	0.672	32
3	0.167	21
4	2.288	33
..	...	...
763	0.171	63
764	0.340	27
765	0.245	30

```
766    0.349    47
767    0.315    23
```

```
[768 rows x 8 columns]
```

```
[10]: y = df['Outcome']
      y
```

```
[10]: 0      1
      1      0
      2      1
      3      0
      4      1
      ..
     763      0
     764      0
     765      0
     766      1
     767      0
      Name: Outcome, Length: 768, dtype: int64
```

```
[11]: print("Features shape:", X.shape)
      print("Labels shape:", y.shape)
      print(y.value_counts())
```

```
Features shape: (768, 8)
Labels shape: (768,)
Outcome
0      500
1      268
      Name: count, dtype: int64
```

```
[12]: X_train, X_test, y_train, y_test = train_test_split(
      X, y, test_size=0.25, random_state=42, stratify=y
      )
```

```
[13]: print(X_train.shape, X_test.shape)
```

```
(576, 8) (192, 8)
```

```
[14]: knn = KNeighborsClassifier(n_neighbors=5)
      knn.fit(X_train, y_train)
      knn_pred = knn.predict(X_test)
```

```
[15]: from sklearn.metrics import (
      confusion_matrix, accuracy_score,
      precision_score, recall_score
```

```
)
```

```
[16]: cm = confusion_matrix(y_test, knn_pred)
      accuracy = accuracy_score(y_test, knn_pred)
      precision = precision_score(y_test, knn_pred)
      recall = recall_score(y_test, knn_pred)
      error_rate = 1 - accuracy
```

```
[17]: print("Confusion Matrix:\n", cm)
```

Confusion Matrix:

```
[[101  24]
 [ 35  32]]
```

```
[18]: print("Accuracy:", accuracy)
```

Accuracy: 0.6927083333333334

```
[19]: print("Error Rate:", error_rate)
```

Error Rate: 0.30729166666666663

```
[20]: print("Precision:", precision)
```

Precision: 0.5714285714285714

```
[21]: print("Recall:", recall)
```

Recall: 0.47761194029850745

```
[ ]:
```